Q6: image enhancement, intensity transformation

#revision

>Define image enhancement and write its purpose.

>Describe intensity transformation using log and gamma transformations and discuss the effects of changing their parameters.

Define image enhancement and write its purpose.

Image Enhancement is the process of manipulating an image so that the result is more suitable than the original for a specific application.

Purpose:

1. **Bring out details:** Highlighting hidden features (like making a dark X-ray readable).
2. **Better Contrast:** Improving visual quality so it is easier for a **human to interpret**.
3. **Pre-processing:** Preparing the image for further scientific analysis (like satellite data).

Describe intensity transformation using log and gamma transformations and discuss the effects of changing their parameters.

Log Transformation:

Definition:

The Log Transformation stretches the dark parts of an image to show hidden details, while squeezing the bright parts together.

The Formula:

$$s = c\log(1 + r)$$

- r: Input pixel intensity.
- s: Output pixel intensity.
- c: A scaling constant.

Effects & Parameters:

- **Stretching :** It stretches the dark pixel values to show hidden details.
- **Compressing:** It squeezes the bright pixel values together.
- **Constant (C):** Acts as a scaling factor to adjust the overall output brightness.

Gamma (Power-Law) Transformation:

Definition: Gamma transformation is a flexible tool used to adjust an image's brightness and contrast by changing a single value called γ (Gamma).

Mathematical Formula:

$$s = c \cdot r^{\gamma}$$

- Where γ (Gamma) is the key parameter.

Effects of Changing Gamma (γ):

- **Case 1:** $\gamma < 1$ **(Brightening)**
  - The curve "bows up."
  - It expands dark values and compresses bright values.
  - **Result:** The image becomes brighter.
- **Case 2:** $\gamma > 1$ **(Darkening)**
  - The curve "bows down."
  - It compresses dark values and expands bright values.
  - **Result:** The image becomes darker with higher contrast.
- **Case 3:** $\gamma = 1$
  - Identity transformation (output equals input).